Plotraa handover: still, blurred front view for live listings

19 Oct 2026 | For: Sant (review, merge, deploy) | From: Pankaj's session with Claude

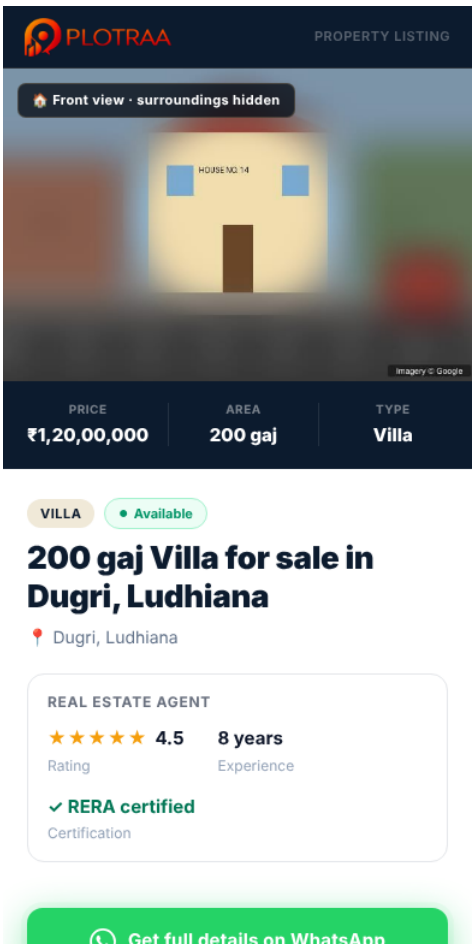
PR	Repo	What	Merge
#60	plotra-backend	Server makes the still blurred photo; live API stops sending exact location	1st
#43	plotra-frontend	Live page shows that photo instead of Street View	2nd (needs #60)

Why

The 18 Oct blur (frontend #41) only placed a blur layer over the live Street View. On a live listing, buyers could still rotate and move the view and see the street through the clear circle. This change replaces it.

What buyers see now (live listings)

- **One still photo** of the property's front, taken from the street and pointed at the house (narrow field of view). No Street View, no rotating, no moving, no zoom.
- **Everything except the house is blurred and dimmed**: neighbouring houses, road, vehicles, signboards.
- Made on the **server** (`src/services/frontViewService.js`, using `sharp`): Google Street View Static API, then blur all but a soft oval on the house, then JPEG. The browser only ever receives the blurred JPEG.
- The live listing API (`GET /public/listings/:slug`) **no longer sends exact lat/lng or the raw satellite/street image URLs** (those also contained our Google API key). It sends `frontViewUrl` instead.
- **Before approval nothing changes**: the dealer still gets the movable satellite and Street View to check and drag the pin.
- No Street View coverage near the property: the section is simply hidden.



Live listing page at phone width, rendered with a test photo: one fixed front view, only the house sharp; neighbours, road and number plate unreadable.

Code changes

Where	Change
Backend (new)	`src/services/frontViewService.js`; route `GET /api/v1/public/listings/:slug/front-view.jpg` (live listings only, 404 otherwise); `test/frontViewService.test.js`
Backend (changed)	`publicListingController.js` (no coords/raw image URLs for live listings, adds frontViewUrl), `routes/public.js`, `package.json` (+ sharp)
Frontend	`PropertyView.jsx`: live listings show the front-view image; Street View only before approval. `PropertyMapMedia.jsx` restored to its pre-#41 version (overlay code removed).

Deploy

- Merge **#60**, then on EC2:

```
git pull && npm ci && pm2 restart all --update-env
```

- `npm ci` is required: it installs `sharp` (image blur). **No migration.**
- Merge **#43**, rebuild and deploy the frontend.
- Street View Static API must be enabled on `GOOGLE_MAPS_API_KEY` (it already is; the old static street image used it).
- Optional env: `FRONT_VIEW_FOV` (default 50; lower = tighter on the house).

Cost

2 Google calls (metadata + 1 static image) per listing per 6 hours per server process, thanks to an in-memory cache. Repeat views in that window cost nothing.

Verified in the sandbox

- Unit tests: **116/116** pass. Blur measured on a test image (centre sharp; edges, road and top blurred), camera heading, one fixed frame with narrow field of view, caching, no imagery returns nothing.
- Against Postgres: live listing API has no coordinates, no image URLs and no API key, and has frontViewUrl; preview listing unchanged; endpoint returns a 640x420 JPEG with cache headers; second view served from cache; non-live or unknown listing returns 404.
- Frontend build passes; live page rendered in headless Chromium at phone width: still photo shown, **zero calls to Google Maps from the browser**.

Check after deploy (real phone)

- Open 5 to 10 live listings on iPhone and Android: one still photo, the house in the sharp area, nothing can be dragged or rotated.
- Open a listing before approval (dealer preview): satellite + Street View still work and the pin can be dragged.
- View page source / network tab of a live listing: no lat/lng and no maps.googleapis.com links.

Known limits and open points

- **Not seen on a real Google photo yet.** Google's camera stands on the nearest road, so in some photos the house may sit a little off-centre and part of it fall in the blur. Tune `FRONT_VIEW_FOV` or the oval (FOCUS in frontViewService.js).
- The house itself, including any house number painted on it, stays visible by design.
- **Google terms:** Google Maps Platform terms restrict modifying and caching Google imagery. Blurring parts of the photo and the 6-hour cache may not be fully compliant; check before scaling. The safe alternative is using the dealer's own front photo.